Discrete Mathematical Structures

MIDSEM PRACTICE PAPER | 30 marks | Suggested time: 90 minutes

Modules 1-3

PROVISIONAL mapping: no DMS channel is visible. Source CO1-CO3 identify lattices/Boolean algebra, combinatorics and graph theory, assumed here to be modules 1-3. Verify against your current outline. Q9 is assigned 4 marks instead of the source 3. Index filenames say MAT 2125; the actual paper header says MAT 2101, Discrete Mathematical Structures.

Answer all questions. Show working and draw labelled diagrams where appropriate. Marks and duration are a practice format, not a confirmed university blueprint.

Question 1 [4 marks]

Draw the Hasse diagram for A = {1, 2, 3, 12, 18, 36}, ordered by divisibility: a is related to b iff a divides b. Is it a lattice? Justify. List all antichains of length more than one.

Source: DMS-R | original Q1A

Question 2 [3 marks]

In a Boolean algebra (A, ∨, ∧, complement), show that:
(i) Every element has a unique complement.
(ii) $a \vee (a' \wedge b) = a \vee b$ and $a \wedge (a' \vee b) = a \wedge b$, for all a,b in A.

Source: DMS-R | original Q1B

Question 3 [3 marks]

Let $E(x_1,x_2,x_3) = (x_1' \wedge x_2 \wedge x_3') \vee (x_1 \wedge x_2') \vee (x_1 \wedge x_3)$ be a Boolean expression over the two-valued Boolean algebra. Write E in both CNF and DNF.

Source: DMS-R | original Q1C

Question 4 [4 marks]

How many ways are there to select a total of 15 items from three types, subject to these limits?
Type A: 1 to 4 items; Type B: 3 to 7 items; Type C: 5 to 10 items.

Source: DMS-R | original Q2A

Question 5 [3 marks]

Show that the number of partitions of n with no part smaller than 3 equals the number of partitions of n+3 whose three largest parts are consecutive integers.

Source: DMS-R | original Q2B

Question 6 [3 marks]

Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order.

Source: DMS-R | original Q2C

Question 7 [3 marks]

Show that if G is a regular self-complementary graph of order n, then $n = 4t + 1$ for some integer t.

Source: DMS-R | original Q3B

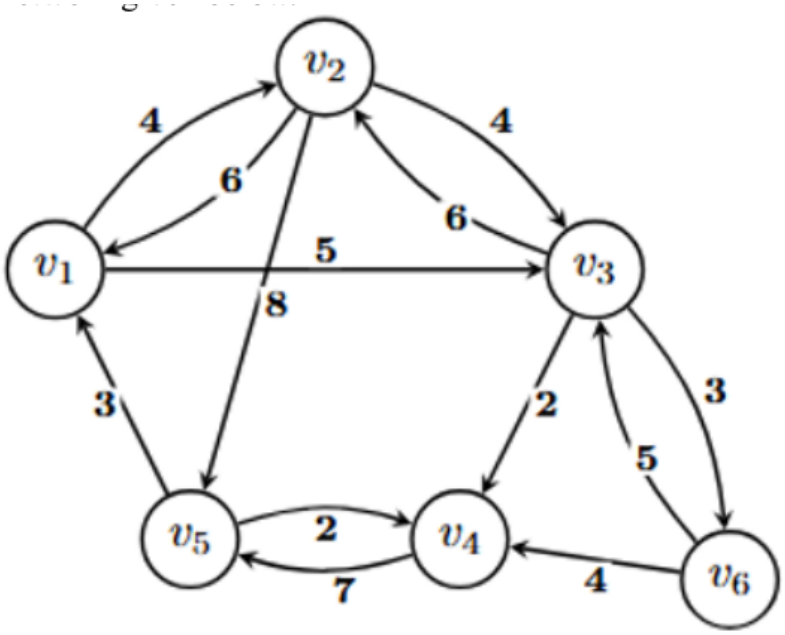
Question 8 [3 marks]

If G has p vertices and minimum degree $\delta(G) \geq (p - 1)/2$, prove that G is connected.

Source: DMS-R | original Q4B

Question 9 [4 marks]

Use Dijkstra's algorithm to find the shortest paths from v_1 to every other vertex in the directed network below. Show the steps.



Source: DMS-R | original Q4C

Sources and selection notes

Working interpretation: NSM means end-semester. This has not yet been confirmed. Questions come only from the local semester-3 CSE papers, using the exact current subject titles. R = regular; M = makeup. Wording is lightly edited, tables transcribed, original question references retained. No new question scenarios were invented.

DMS-R: 2025-26 Regular - MAT_2125-ENGINEERING MATHEMATICS-III

Local source: /Users/shritaake/Desktop/code/paper/public/authoritative/MIT 2021-26/MIT 2025-26 QPs 3rd March 2026/November and December 2025 (Regular)/MIT Sem 1 to Sem 7 QPs/Semester 3/MAT_2125-ENGINEERING MATHEMATICS-III.pdf

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